Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 30 minutes

🪙 Coin-Toss Mendel

You will learn: Random joining of gametes from Tt × Tt parents gives a 1 : 2 : 1 genotype and 3 : 1 phenotype ratio.

YOU NEED

- Two coins (mark Heads = T, Tails = t)
- Tally sheet

STEPS

1. Each coin is one parent's gamete: T (tall allele) or t (dwarf allele).
2. Toss both coins together 100 times.
3. Record TT (both heads), Tt (one each) or tt (both tails) with tally marks.
4. Count each genotype and calculate the ratio.
5. Combine your class data and see whether it moves closer to 1 : 2 : 1.

OBSERVATION TABLE

Offspring	Tally	Count	Expected (out of 100)

QUESTIONS

1. What phenotype ratio did you get?

2. Why does the class total come closer to 1 : 2 : 1?

3. Which law of Mendel does this show?

WHAT IS HAPPENING? — THE SCIENCE

Each parent gives one allele with equal chance — Mendel's law of segregation. The chances combine: TT ¼, Tt ½, tt ¼. Tall is dominant, so TT and Tt look tall: phenotype ratio 3 tall : 1 dwarf. Larger samples get closer to the expected ratio.

Key terms: law of segregation · genotype · phenotype · monohybrid cross

Activity 2 · Hands-on · 🏠 At home · ⏱ 1 hour

🌳 Family Pedigree Chart

You will learn: Pedigree analysis traces a trait through generations using standard symbols.

YOU NEED

- Paper and pen
- Family members willing to take part

⚠️ **Safety:** Ask permission before noting anyone's traits. Do not share family charts publicly.

STEPS

1. Choose an easy-to-see trait such as free or attached earlobes.
2. Record the trait for grandparents, parents, uncles, aunts, siblings and yourself.
3. Draw the pedigree: squares = males, circles = females, shaded = trait present, a horizontal line = marriage, a vertical line = children.
4. Look for a pattern: does the trait skip a generation?
5. Write a possible genotype for each person.

OBSERVATION TABLE

Family member	Male / female	Trait present?	Possible genotype

QUESTIONS

1. Draw the symbols for an affected male and an unaffected female.

2. If a trait appears in a child but not in either parent, is it dominant or recessive?

3. Name one disorder studied using pedigrees.

WHAT IS HAPPENING? — THE SCIENCE

Pedigree analysis is how geneticists study human inheritance, since controlled crosses are not possible. Traits like earlobe attachment are popular for practice, though their real inheritance can be more complex than a single gene.

Key terms: **pedigree analysis** **dominant** **recessive** **autosomal**

Digital activity

💻 Punnett Square Lab

> Choose the genotype of each parent and see the Punnett square and ratios.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch4/2**

> **You will learn:** A Punnett square shows all gamete combinations; dominance decides how genotypes appear as phenotypes.

THINK & WRITE

1. Which cross gives a 1 : 1 ratio? What is it called?

2. What ratio do you get from Tt × Tt?